

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

POSTER TEAM

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Backend Claim

DenseMatrix stores entries in row-major arrays, so matrix code can execute over contiguous data instead of repeatedly re-evaluating pointwise functions.

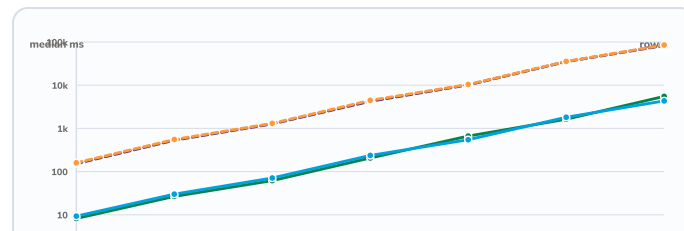


Measures materialized executable workloads, not every `Matrix` expression.

Multiplication Scaling

15.1-21.5x

DenseMatrix vs Matrix → Dense, median ms on a log scale.



Run Gate

2247
RECORDS

328
MATERIALIZED
ROWS

n=2048
MAX ROWS

Poster gate	30 repeats · 5 warmups
Sizes	16, 32, 64, 96, 128, 192, 256
Stress	384, 512, 768, 1024
Machine	pinned core · environment archived

Proof Surface

Representation Tradeoff

Pointwise primitives at the largest complete

Algorithm Evidence

Dense prebuilt Rat workloads, when